

6. Energy-based Model

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MTH345: Deep Generative Models: A Statistical Perspective



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Lecture Topics

M1. Foundations of Probabilistic Modelling

1. Fundamentals of Probability Theory
2. Random Variables
3. Statistical Estimation and Hypothesis Testing
4. Basics of Probabilistic Modelling

M2. Likelihood Ratio-based Generative Models

5. Autoregressive Model
6. Energy-based Model
7. Variational Autoencoder
8. Generative Adversarial Network

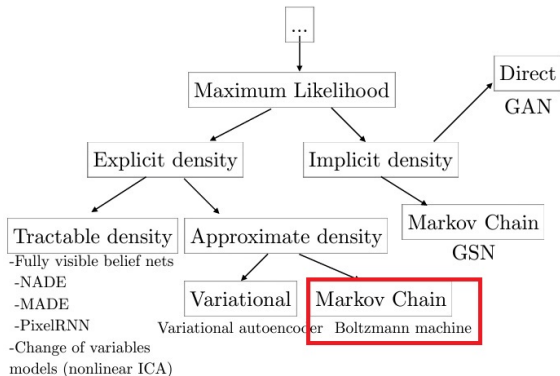
M3. Generative Models with Other Statistical Distance

9. Integral Probability Metric-based Approaches
10. Wasserstein Distance-based Approaches
11. Score-based Approaches

Lecture Topics

- ▶ In the previous chapters, we reviewed Independent Component Analysis (ICA) and autoregressive models.
- ▶ ICA enjoys fast data generation and dimensionality reduction capabilities, but it may lack the ability to express complex data-generating processes.
- ▶ In contrast, autoregressive models can flexibly model complex joint structures, but their sequential generation process is time-consuming.
- ▶ In this lecture, we will review *energy-based models*, another class of latent variable-based generative models.
- ▶ Under certain conditions, these models can universally approximate arbitrary distributions and offer faster data generation than autoregressive models.

Lecture Topics



► Recommended reference materials include:

- Hinton (2002), "Training products of experts by minimizing contrastive divergence," *Neural computation*.
- Goodfellow (2016), "Nips 2016 tutorial: Generative adversarial networks," *arXiv*.

The figure is edited from Goodfellow (2016).

Table of Contents

1. Restricted Boltzmann Machine

2. Temporal Restricted Boltzmann Machine

Limitations of Bayesian Networks



The images are from the Street View House Numbers (SVHN) dataset (Netzer et al., 2011).

Limitations of Bayesian Networks



- ▶ Let us assume, as part of the historical background, that modern houses have progressively transitioned to using blue-backgrounded signs.
- ▶ Let Z_1 indicate whether the house has an administrative origin (e.g., conforming to modern building codes), and Z_2 indicate whether the digit is the ending one.
- ▶ Given $\mathbf{x}$, Z_1 and Z_2 naturally preserve (conditional) independence, i.e.,

$$Z_1 \perp\!\!\!\perp Z_2 \mid \mathbf{x}. \quad (1)$$

Limitations of Bayesian Networks

- ▶ Many latent variable models, including ICA, are formulated as Bayesian networks that satisfy

$$f(\mathbf{z}) = \prod_{j=1}^r f(z_j) \quad \text{and} \quad f_{\theta}(\mathbf{x} \mid \mathbf{z}) = \prod_{j=1}^d f_{\theta}(x_j \mid \mathbf{z}). \quad (2)$$

- ▶ Although the prior latent components $\mathbf{z}$ are statistically independent, after conditioning on the observed data $\mathbf{x}$, the posterior ones can become dependent:

$$f_{\theta}(\mathbf{z} \mid \mathbf{x}) = \frac{\prod_{j=1}^d f_{\theta}(x_j \mid \mathbf{z}) \prod_{j=1}^r f(z_j)}{f_{\theta}(\mathbf{x})} \neq \prod_{j=1}^r f_{\theta}(z_j \mid \mathbf{x}) \quad (3)$$

This phenomenon is known as *explaining away* in directed graphical models.

Energy-based Models

- ▶ Restricted Boltzmann machines (RBMs) are latent variable models that guarantee conditional independence among variables within the same layer.
- ▶ We first review energy-based models and general Boltzmann machines.
- ▶ *Energy-based models* are generative models that utilize an energy function to define a *Boltzmann distribution*¹:

$$f_{\theta}(\mathbf{x}) := \frac{\exp(-E_{\theta}(\mathbf{x}))}{C(\theta)} \quad (4)$$

where $E_{\theta}(\mathbf{x})$ denotes the energy function, and $C(\theta) := \int \exp(-E_{\theta}(\mathbf{x}))d\mathbf{x}$ is the partition function (or normalization constant).

¹Here, we assume unit temperature and a unit Boltzmann constant.

Energy-based Models: Gaussian Example

- ▶ For example, the multivariate Gaussian distribution with parameters $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is given by

$$f_{\theta}(\mathbf{x}) = (2\pi)^{-d/2} |\boldsymbol{\Sigma}|^{-1/2} \exp \left(-(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) / 2 \right), \quad (5)$$

which yields the energy function

$$E_{\theta}(\mathbf{x}) = \frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}). \quad (6)$$

- ▶ Its corresponding partition function (or normalization constant) is $C(\theta) = (2\pi)^{d/2} |\boldsymbol{\Sigma}|^{1/2}$.²

²In general, any distribution in the exponential family, represented by $p_{\theta}(\mathbf{x}) = h(\mathbf{x}) \exp(\boldsymbol{\eta}(\theta)^T \mathbf{T}(\mathbf{x}) - A(\theta))$, can be interpreted as an instance of Equation (4). In this case, the energy function is $E_{\theta}(\mathbf{x}) = -\boldsymbol{\eta}(\theta)^T \mathbf{T}(\mathbf{x}) - \log h(\mathbf{x})$ and the partition function is $C(\theta) = \exp(A(\theta))$.

Boltzmann Machine

- ▶ Boltzmann machines (Ackley et al., 1985) are classical energy-based models that introduce latent variables to define a joint energy density function over both the latent variables and observations.
- ▶ The energy function of a Boltzmann machine is expressed as

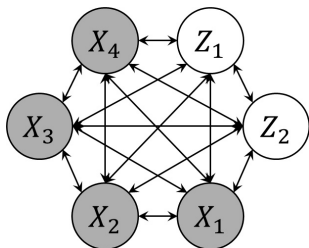
$$E_{\theta}(\mathbf{z}, \mathbf{x}) := - \left(\frac{1}{2} \begin{pmatrix} \mathbf{z} \\ \mathbf{x} \end{pmatrix}^T \mathbf{W} \begin{pmatrix} \mathbf{z} \\ \mathbf{x} \end{pmatrix} + \begin{pmatrix} \boldsymbol{\alpha} \\ \boldsymbol{\beta} \end{pmatrix}^T \begin{pmatrix} \mathbf{z} \\ \mathbf{x} \end{pmatrix} \right), \quad (7)$$

where $\theta := (\mathbf{W}, \boldsymbol{\alpha}, \boldsymbol{\beta})$ represents the parameters. Here, $\mathbf{W}$ is a symmetric matrix with zero diagonal elements, indicating the absence of self-loops.

- ▶ The corresponding joint density is then given by

$$p_{\theta}(\mathbf{z}, \mathbf{x}) \propto \exp \left(\frac{1}{2} \begin{pmatrix} \mathbf{z} \\ \mathbf{x} \end{pmatrix}^T \begin{pmatrix} \mathbf{W}_{ZZ} & \mathbf{W}_{ZX} \\ \mathbf{W}_{ZX}^T & \mathbf{W}_{XX} \end{pmatrix} \begin{pmatrix} \mathbf{z} \\ \mathbf{x} \end{pmatrix} + \begin{pmatrix} \boldsymbol{\alpha} \\ \boldsymbol{\beta} \end{pmatrix}^T \begin{pmatrix} \mathbf{z} \\ \mathbf{x} \end{pmatrix} \right). \quad (8)$$

Boltzmann Machine



- For example, when we have $\mathbf{x} \in \{0, 1\}^{d=4}$ and $\mathbf{z} \in \{0, 1\}^{r=2}$, the joint density function can be explicitly expanded as:

$$\begin{aligned}
 p_{\theta}(\mathbf{z}, \mathbf{x}) \propto \exp \bigg(& W_{ZZ,1,2} z_1 z_2 + \sum_{i=1}^2 \sum_{j=1}^4 W_{ZX,i,j} z_i x_j \\
 & + \sum_{j < k} W_{XX,j,k} x_j x_k + \sum_{i=1}^2 \alpha_i z_i + \sum_{j=1}^4 \beta_j x_j \bigg).
 \end{aligned} \tag{9}$$

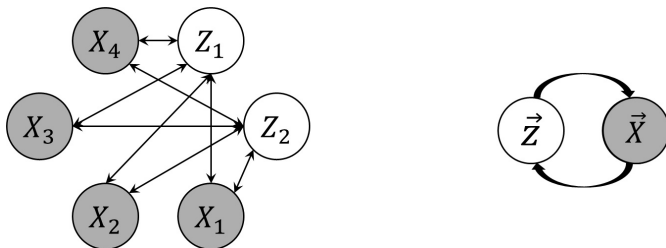
Boltzmann Machine

- ▶ Boltzmann machines assume binary latent variables and binary observed features.
- ▶ The tractable joint energy forms provide an explicit interpretation for logit functions. For a given $\mathbf{x}_{-j} := (x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_d)^T$, let $\mathbf{x}_{j,\text{on}} := (x_1, \dots, x_j = 1, \dots, x_d)^T$ and $\mathbf{x}_{j,\text{off}} := (x_1, \dots, x_j = 0, \dots, x_d)^T$. We then have

$$\begin{aligned} \log \left(\frac{p_{\theta}(\mathbf{z}, \mathbf{x}_{j,\text{on}})}{p_{\theta}(\mathbf{z}, \mathbf{x}_{j,\text{off}})} \right) &= -E_{\theta}(\mathbf{z}, \mathbf{x}_{j,\text{on}}) + E_{\theta}(\mathbf{z}, \mathbf{x}_{j,\text{off}}) \\ &= (\mathbf{W}_{ZX}^T \mathbf{z} + \mathbf{W}_{XX} \mathbf{x} + \beta)_j. \end{aligned} \quad (10)$$

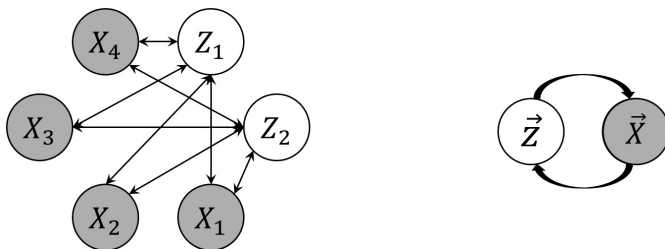
- ▶ However, due to the intra-layer connections within $\mathbf{z}$ and $\mathbf{x}$, represented respectively by $\mathbf{W}_{ZZ}$ and $\mathbf{W}_{XX}$, it remains computationally intractable to evaluate the conditional posteriors $p_{\theta}(\mathbf{z} | \mathbf{x})$, $p_{\theta}(\mathbf{x} | \mathbf{z})$, and their corresponding logits.

Restricted Boltzmann Machine



- ▶ Restricted Boltzmann machines (RBMs) simplify general Boltzmann machines by eliminating connections within latent variables and within observed features, thereby defining a bipartite graph.
- ▶ They were originally proposed by Smolensky et al. (1986) under the name *Harmonium* and gained widespread popularity after Hinton (2002) introduced efficient training algorithms.

Restricted Boltzmann Machine



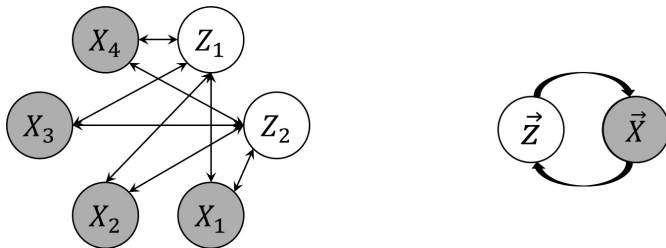
- The energy function of an RBM is explicitly expressed as

$$E_{\theta}(\mathbf{z}, \mathbf{x}) := -(\mathbf{z}^T \mathbf{W} \mathbf{x} + \boldsymbol{\alpha}^T \mathbf{z} + \boldsymbol{\beta}^T \mathbf{x}), \quad (11)$$

which directly corresponds to Equation (8) under the structural constraints $\mathbf{W}_{ZZ} = \mathbf{0}$, $\mathbf{W}_{XX} = \mathbf{0}$, and $\mathbf{W}_{ZX} = \mathbf{W}$.³

³RBM's traditionally assume binary observations and binary latent variables, but they can be extended to generic exponential family distributions (Welling et al., 2004).

Restricted Boltzmann Machine



- For example, when we have $\mathbf{x} \in \{0, 1\}^{d=4}$ and $\mathbf{z} \in \{0, 1\}^{r=2}$, the joint density $p_{\theta}(\mathbf{z}, \mathbf{x})$ simplifies to:

$$p_{\theta}(\mathbf{z}, \mathbf{x}) \propto \exp \left(\sum_{i=1}^2 \sum_{j=1}^4 W_{i,j} z_i x_j + \sum_{i=1}^2 \alpha_i z_i + \sum_{j=1}^4 \beta_j x_j \right). \quad (12)$$

Restricted Boltzmann Machine

- ▶ The joint density $p_{\theta}(\mathbf{z}, \mathbf{x})$ exhibits an exact conditional independence property:

$$p_{\theta}(\mathbf{x} | \mathbf{z}) = \prod_{j=1}^d p_{\theta}(x_j | \mathbf{z}) \quad \text{and} \quad p_{\theta}(\mathbf{z} | \mathbf{x}) = \prod_{i=1}^r p_{\theta}(z_i | \mathbf{x}). \quad (13)$$

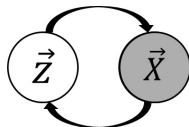
- ▶ Furthermore, RBMs provide analytically tractable inference for each node:

$$p_{\theta}(x_j = 1 | \mathbf{z}) = \text{sig}((\mathbf{W}^T \mathbf{z} + \beta)_j) \quad \text{and} \quad p_{\theta}(z_i = 1 | \mathbf{x}) = \text{sig}((\mathbf{W} \mathbf{x} + \alpha)_i). \quad (14)$$

- ▶ Consequently, RBMs have linear conditional logit functions:

$$\log \left(\frac{p_{\theta}(z_1, \dots, z_i = 1, \dots, z_r | \mathbf{x})}{p_{\theta}(z_1, \dots, z_i = 0, \dots, z_r | \mathbf{x})} \right) = (\mathbf{W} \mathbf{x} + \alpha)_i. \quad (15)$$

Restricted Boltzmann Machine



- RBMs generate synthetic data via the following M -step Gibbs sampling process:
1. Initialize by sampling $\tilde{\mathbf{X}}_1$ from a pre-specified distribution $p(\tilde{\mathbf{x}})^4$ and draw $\tilde{\mathbf{Z}}_1 \sim p_\theta(\mathbf{z} \mid \mathbf{x} = \tilde{\mathbf{X}}_1)$.
 2. For $m = 1, \dots, M-1$, sequentially generate $\tilde{\mathbf{X}}_{m+1} \sim p_\theta(\mathbf{x} \mid \mathbf{z} = \tilde{\mathbf{Z}}_m)$ and $\tilde{\mathbf{Z}}_{m+1} \sim p_\theta(\mathbf{z} \mid \mathbf{x} = \tilde{\mathbf{X}}_{m+1})$.
 3. The resulting pair $(\tilde{\mathbf{Z}}_M, \tilde{\mathbf{X}}_M)$ constitutes the M -step Gibbs sample. Under certain regularity conditions, the joint density of $(\tilde{\mathbf{Z}}_M, \tilde{\mathbf{X}}_M)$ converges to the true joint distribution $p_\theta(\mathbf{z}, \mathbf{x})$ as $M \rightarrow \infty$.

⁴A common choice is to utilize the empirical distribution of real observations $\mathbf{X}$.

Contrastive Learning

- ▶ (Restricted) Boltzmann machines aim to find maximum likelihood estimators. However, the log-likelihood function,

$$\log p_{\theta}(\mathbf{x}) = -\log C(\theta) + \log \sum_{\mathbf{z}} \exp(-E_{\theta}(\mathbf{z}, \mathbf{x})), \quad (16)$$

contains the intractable $C(\theta)$, posing a severe computational challenge.

- ▶ Contrastive divergence (CD) (Hinton, 2002) circumvents this issue by evaluating the gradient of the log-likelihood (the score function) directly:

$$\nabla_{\theta} \log p_{\theta}(\mathbf{x}) = - \underbrace{\sum_{\mathbf{z}} \left(\nabla_{\theta} E_{\theta}(\mathbf{z}, \mathbf{x}) \right) p_{\theta}(\mathbf{z} | \mathbf{x})}_{\text{Positive gradient}} + \underbrace{\sum_{\mathbf{x}, \mathbf{z}} \left(\nabla_{\theta} E_{\theta}(\mathbf{z}, \mathbf{x}) \right) p_{\theta}(\mathbf{z}, \mathbf{x})}_{\text{Negative gradient}}. \quad (17)$$

- ▶ The score function is exactly the contrast between the positive and negative phases, which is the direct origin of the name *contrastive divergence*.

Contrastive Learning

- ▶ Driven by this formulation, the CD algorithm applies short-run Gibbs sampling to approximate the intractable expectations in both phases.
- ▶ For a given training dataset $(\mathbf{x}_i)_{i=1}^n$, the execution of the CD algorithm is formalized as follows:
 1. **Positive Phase:** For a given training observation $\mathbf{x}_i$, sample $\tilde{\mathbf{z}}_i \sim p_{\theta}(\mathbf{z} \mid \mathbf{x} = \mathbf{x}_i)$ to construct the i -th *positive sample* $(\tilde{\mathbf{z}}_i, \mathbf{x}_i)$.
 2. **Negative Phase:** Starting from $\mathbf{x}_i$, draw the i -th *negative sample* $(\tilde{\mathbf{z}}_{i,M}, \tilde{\mathbf{x}}_{i,M})$ using M -step Gibbs sampling.
 3. **Gradient Evaluation:** Compute the empirical positive and negative gradients using the sampled mini-batches.
 4. **Parameter Update:** Update θ by ascending the approximated log-likelihood gradient.

Generation Result

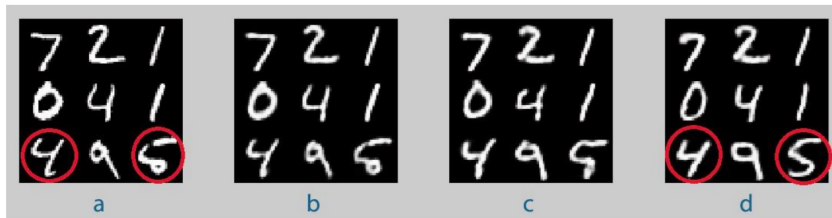


Figure 8: Results from an RBM with 250 hidden units that has been trained on MNIST dataset. In this experiment, after extracting 250 features from 9 MNIST images (28*28 pixel), the new images have been generated from extracted features. (a) 9 MNIST images. (b) Generated images from extracted features with $k = 1$ sampling iteration. (c) Generated images from extracted features with $k = 10$ sampling iterations. (d) Generated images from extracted features with $k = 100$ sampling iterations. The related code is in “*test_generateData.m*”.

Images are from Keyvanrad and Homayounpour (2014).

Flexibility vs. Interpretability

- ▶ Compared with general Boltzmann machines, RBMs sacrifice intra-layer flexibility by eliminating connections within $\mathbf{z}$ and within $\mathbf{x}$.
- ▶ Nonetheless, RBMs function as distributional universal approximators:

Theorem (Representational Power of RBMs; Le Roux and Bengio, 2008)

For any $\epsilon > 0$ and any target probability density function $p(\mathbf{x})$ supported on $\{0, 1\}^d$, there exists an RBM parameterized by θ such that

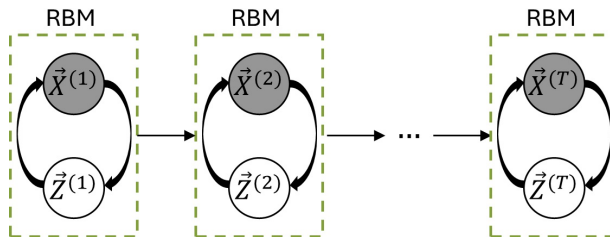
$$D_{\text{KL}}(p(\mathbf{x}) \parallel p_{\theta}(\mathbf{x})) < \epsilon. \quad (18)$$

Table of Contents

1. Restricted Boltzmann Machine

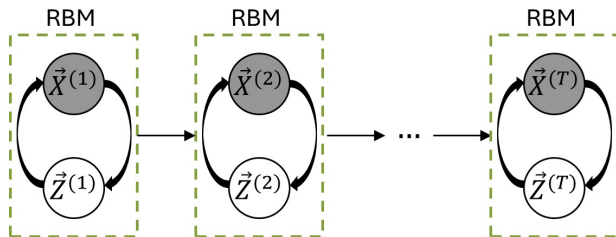
2. Temporal Restricted Boltzmann Machine

Temporal Restricted Boltzmann Machine



- ▶ RBMs face challenges in handling temporal data because they do not permit intra-layer or inter-temporal connections among observations across different time steps.
- ▶ Temporal RBMs (TRBMs) (Taylor et al., 2006) are a temporal extension of RBMs that incorporates directed connections from past latent and observed variables to present ones.

Temporal Restricted Boltzmann Machine

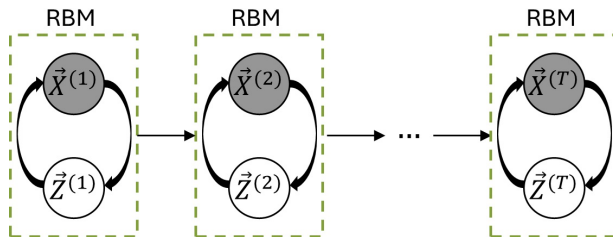


- TRBMs first construct a standard RBM for the initial time step:

$$p_{\theta}(\mathbf{z}^{(1)}, \mathbf{x}^{(1)}) \propto \exp(-E_{\theta}^{(1)}(\mathbf{z}^{(1)}, \mathbf{x}^{(1)})), \quad (19)$$

where $E_{\theta}^{(1)}(\mathbf{z}^{(1)}, \mathbf{x}^{(1)}) := -((\mathbf{z}^{(1)})^T \mathbf{W} \mathbf{x}^{(1)} + \boldsymbol{\alpha}^T \mathbf{z}^{(1)} + \boldsymbol{\beta}^T \mathbf{x}^{(1)})$.

Temporal Restricted Boltzmann Machine



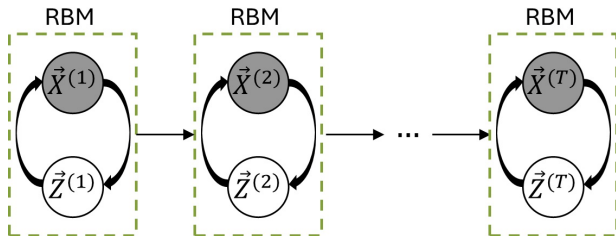
- Next, the conditional distribution of $(\mathbf{z}^{(2)}, \mathbf{x}^{(2)})$ given the initial variables at the first time step can be expressed as:

$$p_{\theta}(\mathbf{z}^{(2)}, \mathbf{x}^{(2)} \mid \mathbf{z}^{(1)}, \mathbf{x}^{(1)}) \propto \exp(-E_{\theta}^{(2)}(\mathbf{z}^{(2)}, \mathbf{x}^{(2)}; \mathbf{z}^{(1)}, \mathbf{x}^{(1)})), \quad (20)$$

where

$$E_{\theta}^{(2)}(\mathbf{z}^{(2)}, \mathbf{x}^{(2)}; \mathbf{z}^{(1)}, \mathbf{x}^{(1)}) := -((\mathbf{z}^{(2)})^T \mathbf{W} \mathbf{x}^{(2)} + (\alpha + \mathbf{U} \mathbf{z}^{(1)})^T \mathbf{z}^{(2)} + \beta^T \mathbf{x}^{(2)}).$$

Temporal Restricted Boltzmann Machine



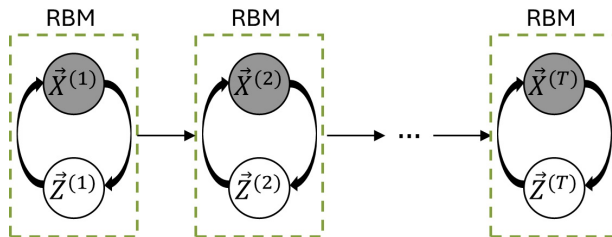
- Finally, TRBMs assume time homogeneity and sequentially stack RBMs:

$$p_{\theta}(\mathbf{z}^{(t)}, \mathbf{x}^{(t)} \mid \mathbf{z}^{(1:(t-1))}, \mathbf{x}^{(1:(t-1))}) \propto \exp \left(-E_{\theta}^{(t)}(\mathbf{z}^{(t)}, \mathbf{x}^{(t)}; \mathbf{z}^{(1:(t-1))}, \mathbf{x}^{(1:(t-1))}) \right) \quad (21)$$

where the energy function is defined as

$$E_{\theta}^{(t)}(\mathbf{z}^{(t)}, \mathbf{x}^{(t)}; \mathbf{z}^{(1:(t-1))}, \mathbf{x}^{(1:(t-1))}) := -((\mathbf{z}^{(t)})^T \mathbf{W} \mathbf{x}^{(t)} + (\alpha + \mathbf{U} \mathbf{z}^{(t-1)})^T \mathbf{z}^{(t)} + \beta^T \mathbf{x}^{(t)}). \quad (22)$$

Temporal Restricted Boltzmann Machine



- ▶ Note that $E_{\theta}^{(t)}(\mathbf{z}^{(t)}, \mathbf{x}^{(t)}; \mathbf{z}^{(1:t-1)}, \mathbf{x}^{(1:t-1)})$ is a function of $(\mathbf{z}^{(t-1)}, \mathbf{z}^{(t)}, \mathbf{x}^{(t)})$. Consequently, $p_{\theta}(\mathbf{z}^{(t)}, \mathbf{x}^{(t)} | \mathbf{z}^{(1:t-1)}, \mathbf{x}^{(1:t-1)}) = p_{\theta}(\mathbf{z}^{(t)}, \mathbf{x}^{(t)} | \mathbf{z}^{(t-1)})$.
- ▶ TRBMs function as distributional universal approximators (Odenne and Edwards, 2016).

Temporal Restricted Boltzmann Machine

- ▶ Since $p_{\theta}(\mathbf{z}^{(t)}, \mathbf{x}^{(t)} \mid \mathbf{z}^{(1:(t-1))}, \mathbf{x}^{(1:(t-1))})$ is formulated as an RBM under the energy function $E_{\theta}^{(t)}$, it inherits the exact conditional independence property:

$$p_{\theta}(\mathbf{x}^{(t)} \mid \mathbf{z}^{(1:t)}, \mathbf{x}^{(1:(t-1))}) = \prod_{j=1}^d p_{\theta}(x_j^{(t)} \mid \mathbf{z}^{(1:t)}, \mathbf{x}^{(1:(t-1))}) \quad (23)$$

and

$$p_{\theta}(\mathbf{z}^{(t)} \mid \mathbf{z}^{(1:(t-1))}, \mathbf{x}^{(1:t)}) = \prod_{i=1}^r p_{\theta}(z_i^{(t)} \mid \mathbf{z}^{(1:(t-1))}, \mathbf{x}^{(1:t)}). \quad (24)$$

Temporal Restricted Boltzmann Machine

- ▶ TRBMs also provide analytically tractable and interpretable inference for each node:

$$p_{\theta}(z_i^{(t)} = 1 \mid \mathbf{z}^{(1:(t-1))}, \mathbf{x}^{(1:t)}) = \text{sig}((\mathbf{W}\mathbf{x}^{(t)} + \alpha + \mathbf{U}\mathbf{z}^{(t-1)})_i). \quad (25)$$

Similarly,

$$p_{\theta}(x_j^{(t)} = 1 \mid \mathbf{z}^{(1:t)}, \mathbf{x}^{(1:(t-1))}) = \text{sig}((\mathbf{W}^T \mathbf{z}^{(t)} + \beta)_j). \quad (26)$$

- ▶ Their corresponding conditional logit functions are strictly linear and can be expressed as:

$$\log \left(\frac{p_{\theta}(z_1^{(t)}, \dots, z_i^{(t)} = 1, \dots, z_r^{(t)} \mid \mathbf{z}^{(1:(t-1))}, \mathbf{x}^{(1:t)})}{p_{\theta}(z_1^{(t)}, \dots, z_i^{(t)} = 0, \dots, z_r^{(t)} \mid \mathbf{z}^{(1:(t-1))}, \mathbf{x}^{(1:t)})} \right) = (\mathbf{W}\mathbf{x}^{(t)} + \alpha + \mathbf{U}\mathbf{z}^{(t-1)})_i. \quad (27)$$

Temporal Restricted Boltzmann Machine

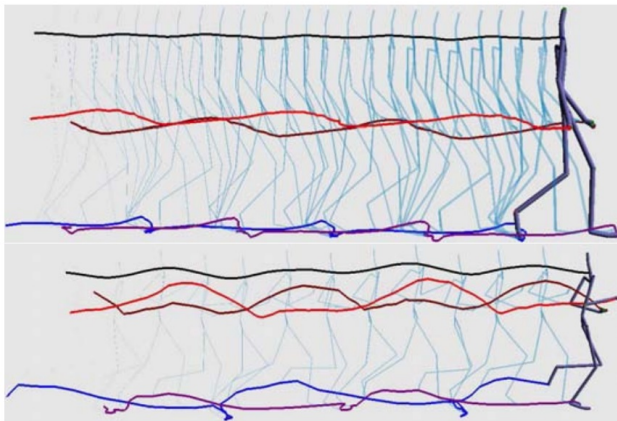
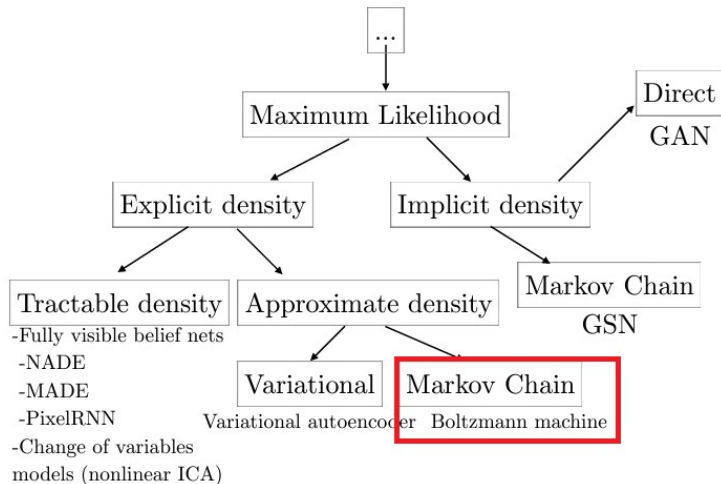


Figure 3: After training, the same model can generate walking (top) and running (bottom) motion (see videos on the website). Each skeleton is 4 frames apart.

Images are from Taylor et al. (2006).

Lecture Topics



The figure is edited from Goodfellow (2016).

References II

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